

DM9051 SPI Fast Ethernet Driver - Troubleshooting Guide

Table of Contents

1. [Troubleshooting](#)
 2. [Examples](#)
-

Overview

The DM9051 driver is a Linux network driver for the Davicom DM9051 SPI Fast Ethernet controller. This guide focuses on configuring and tuning the driver for optimal network performance on different hardware platforms.

Troubleshooting

Low Throughput Issues

Symptoms: Lower than expected bandwidth or packet rate

Diagnosis Steps:

1. Check current configuration:

```
dmesg | grep -i dm9051
```

2. Verify interrupt mode is enabled:
 - Look for "Operation: Interrupt mode" in dmesg
 - Not "Operation: Polling mode"
3. Check SPI transfer mode:
 - Look for "SPI_XFER_MEM= burst mode" (preferred)
 - Not "SPI_XFER_MEM= alignment mode"

Solutions:

- Set to enable `BURST_MODE_FULL` if using alignment mode
- Set to enable interrupt mode if using polling
- Set to enable checksum offload
- Check SPI bus speed which is configured in the device tree

High CPU Usage

Symptoms: CPU usage > 50% for network processing

Diagnosis:

1. Check if polling mode is active
2. Check if checksum offload is disabled
3. Monitor packet rate vs. CPU usage

Solutions:

- Switch to interrupt mode (`MODE_INTERRUPT`)
- Enable checksum offload (`DEFAULT_CHECKSUM_ON`)
- Increase polling delays if polling mode is required
- Check for excessive error handling (monitor error counters)

Packet Loss or Errors

Symptoms: RX/TX errors, dropped packets

Diagnosis:

1. Check error counters:

```
ethtool -S eth0 | grep error
```

2. Enable monitoring:

```
.force_monitor_rxb = FORCE_MONITOR_RXB,  
.force_monitor_tx_timeout = FORCE_MONITOR_TX_TIMEOUT,
```

Solutions:

- Increase TX timeout if timeouts occur
- Check SPI bus stability (reduce speed if needed)
- Switch to alignment mode if burst mode causes errors
- Adjust block sizes for alignment mode

SPI Transfer Errors

Symptoms: SPI transaction failures, regmap errors

Solutions:

- Reduce SPI bus speed in device tree

- Switch from burst mode to alignment mode
- Reduce block sizes in alignment mode
- Verify SPI controller compatibility

Configuration Not Applied

Symptoms: Changes don't take effect

Solution:

1. Rebuild the driver module
2. Unload and reload the module:

```
rmmod dm9051
insmod dm9051
```

3. Check dmesg for configuration messages
4. Verify no compilation errors

Examples

Example 1: High-Performance Configuration for RPi4

```
const struct driver_config confdata = {
    .release_version = "lnx_dm9051_kt6631_r2503_v3.9.1_R2",
    .interrupt = MODE_INTERRUPT, // Use interrupts
    .mid = MODE_B, // RPi4 optimized
    .mod = {
        [MODE_A] = { /* ... */ },
        [MODE_B] = {
            .test_info = "RPi4 High Performance",
            .skb_wb_mode = SKB_WB_ON,
            .tx_mode = FORCE_TX_CONTI_ON, // Enable TX continuation
            .checksumming = DEFAULT_CHECKSUM_ON, // Enable checksum offload
            .align = {
                .burst_mode = BURST_MODE_FULL, // Full burst mode
                .tx_blk = 0,
                .rx_blk = 0,
            }
        },
        [MODE_C] = { /* ... */ }
    }
};
```

Example 2: Custom Alignment Mode Configuration

```
[MODE_A] = {
    .test_info = "Custom Alignment Mode",
    .skb_wb_mode = SKB_WB_ON,
    .tx_mode = FORCE_TX_CONTI_OFF,
    .checksumming = DEFAULT_CHECKSUM_ON,
    .align = {
        .burst_mode = BURST_MODE_ALIGN, // Use alignment mode
        .tx_blk = 64, // 64-byte TX blocks
        .rx_blk = 128, // 128-byte RX blocks
    }
}
```

Example 3: Polling Mode with Custom Delays

```
// In dm9051.c
const struct driver_config confdata = {
    .interrupt = MODE_POLL, // Use polling mode
    // ...
};

// In dm9051.h
const struct eng_config engdata = {
    .force_monitor_rxb = FORCE_SILENCE_RXB,
    .force_monitor_rxc = FORCE_SILENCE_RX_COUNT,
    .force_monitor_tx_timeout = FORCE_SILENCE_TX_TIMEOUT,
    .sched = {
        .delayF = {0, 1, 2, 1, 1}, // Custom polling delays
        .nTargetMaxNum = 5, // Use all 5 delay values
    },
    .tx_timeout_us = 3000, // 3 ms timeout
};
```

Example 4: Debug Configuration

```
[MODE_B] = {
    .test_info = "Debug Mode",
    .skb_wb_mode = SKB_WB_ON,
    .tx_mode = FORCE_TX_CONTI_OFF, // Standard mode for debugging
    .checksumming = DEFAULT_CHECKSUM_OFF, // Software checksum for debugging
    .align = {
        .burst_mode = BURST_MODE_ALIGN, // More compatible
        .tx_blk = 32,
```

```
        .rx_blk = 64,  
    }  
}  
  
// In dm9051.h  
const struct eng_config engdata = {  
    .force_monitor_rxb = FORCE_MONITOR_RXB, // Enable monitoring  
    .force_monitor_rxc = FORCE_MONITOR_RX_COUNT,  
    .force_monitor_tx_timeout = FORCE_MONITOR_TX_TIMEOUT, // Enable timeout  
monitoring  
    // ...  
};
```

Document Version: 1.0

Last Updated: 2025-11-20

Driver Version: r2503_v3.9.1_R2